

CLUSTER ANALYSIS

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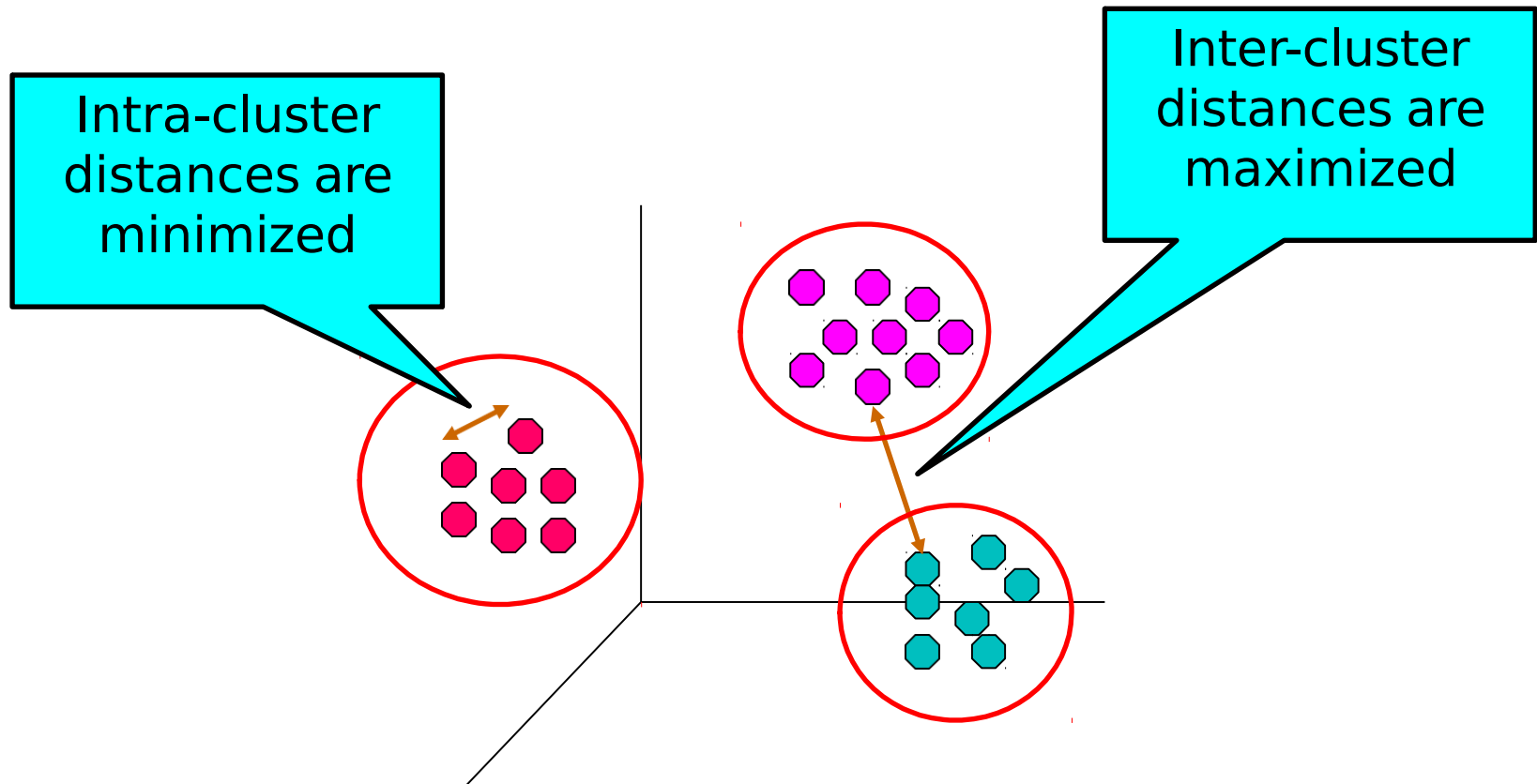
- Divides data into groups (clusters) that are meaningful, useful or both
- Clustering for Understanding
 - Conceptually meaningful groups of objects that share common characteristics
- Clustering for Utility
 - Summarization-Reduce the size of large data sets

What is Cluster Analysis?

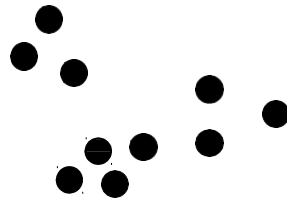
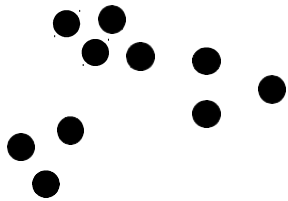
- Groups data objects based only on information found in the data that describes the objects and their relationships
- **Goal** : objects within a group be similar (or related) to one another and different from (or unrelated to) the objects in other groups
- The greater the similarity (or homogeneity) within a group and the greater the difference between groups, the better or more distinct the clustering

What is Cluster Analysis?

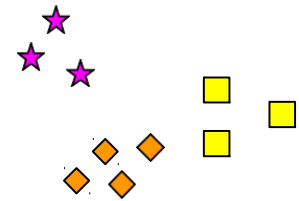
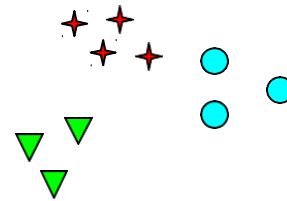
- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



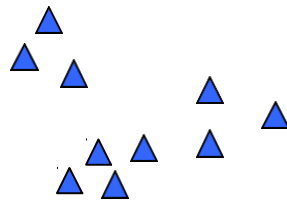
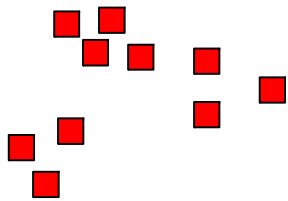
Notion of a Cluster can be Ambiguous



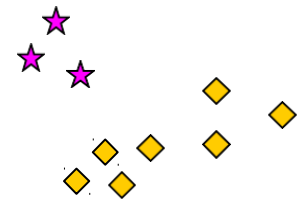
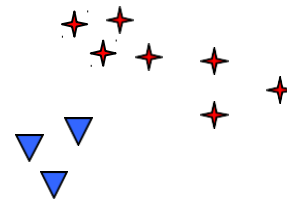
How many clusters?



Six Clusters



Two Clusters

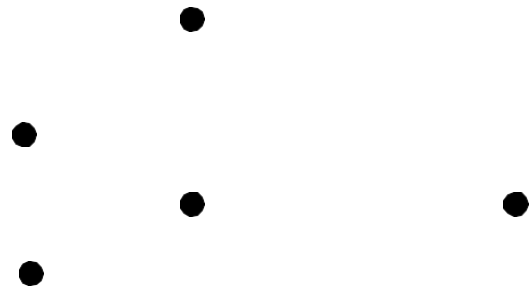
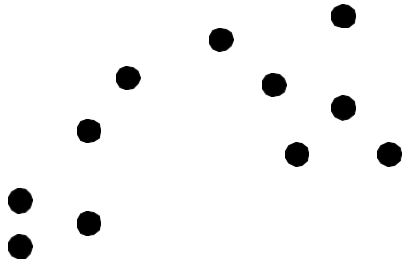


Four Clusters

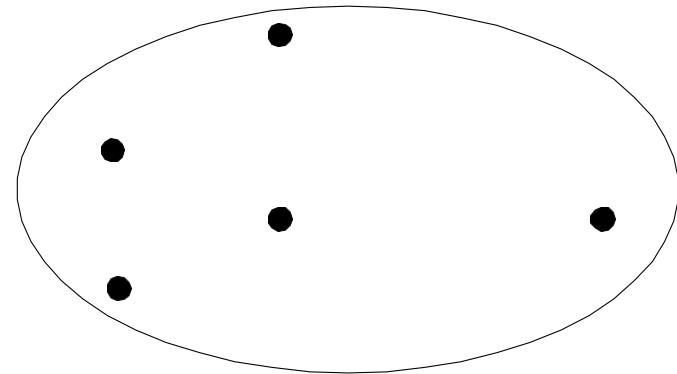
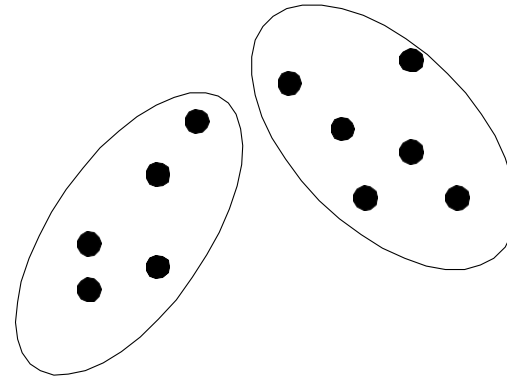
Different Types of Clusterings

- An entire collection of clusters is commonly referred to as a clustering
- **Hierarchical versus Partitional**
- Partitional clustering is simply a division of the set of data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
- If we permit clusters to have subclusters, then we obtain a hierarchical clustering- set of nested clusters that are organized as a tree
- Each node (cluster) in the tree (except for the leaf nodes) is the union of its children (subclusters), and the root of the tree is the cluster containing all the objects

Partitional Clustering

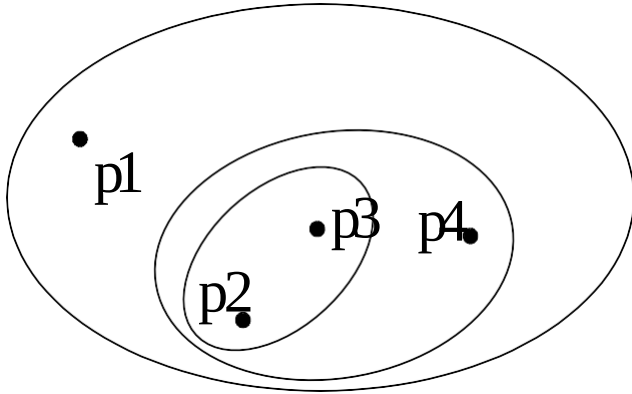


Original Points

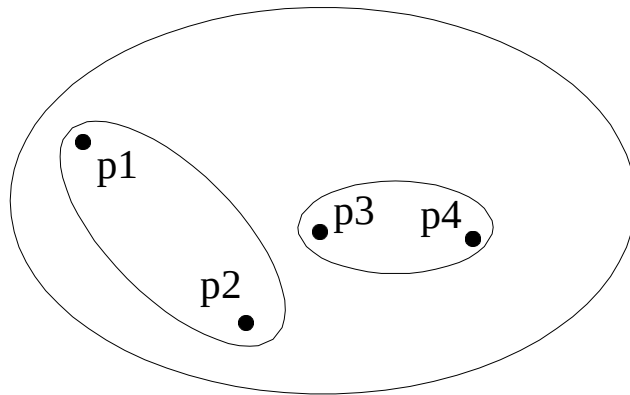


A Partitional Clustering

Hierarchical Clustering



Traditional Hierarchical Clustering



Non-traditional Hierarchical Clustering

Exclusive versus Overlapping versus Fuzzy

- **Exclusive**

- They assign each object to a single cluster

- **Overlapping or non-exclusive clustering**

- An object can simultaneously belong to more than one group (class)
- Also often used when an object is "between" two or more clusters and could reasonably be assigned to any of these cluster



■ Fuzzy clustering

- In a fuzzy clustering, every object belongs to every cluster with a membership weight that is between 0 (absolutely doesn't belong) and 1 (absolutely belongs)
- Sum of the weights for each object must equal 1
- Probabilistic clustering techniques compute the probability with which each point belongs to each cluster, and these probabilities must also sum to 1
- Appropriate for avoiding the arbitrariness of assigning an object to only one cluster when it may be close to several
- Often converted to an exclusive clustering by assigning each object to the cluster in which its membership weight or probability is highest

Complete versus Partial

- A complete clustering assigns every object to a cluster, whereas a partial clustering does not
- Motivation for a partial clustering is that some objects in a data set
 - may not belong to well-defined group
- Many times objects in the data set may represent noise, outliers, or "uninteresting background."
- Example: some newspaper stories may share a common theme, such as global warming, while other stories are more generic or one-of-a-kind. Thus, to find the important topics in last month's stories, we may want to search only for clusters of documents that are tightly related by a common theme

Types of Clusters

- Well-separated clusters
- Prototype-based clusters
- Contiguous clusters
- Density-based clusters
- Property or Conceptual

Types of Clusters: Well-Separated

- Well-Separated Clusters:
 - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.



3 well-separated clusters

Prototype-Based clusters

- A cluster is a set of objects in which each object is closer (more similar) to the prototype that defines the cluster than to the prototype of any other cluster
- For data with continuous attributes, the prototype of a cluster is often a centroid, i.e., the average (mean) of all the points in the cluster
- When a centroid is not meaningful, such as when the data has categorical attributes, the prototype is often a medoid, i.e., the most representative point of a cluster.

Types of Clusters: Center-Based

□ Center-based

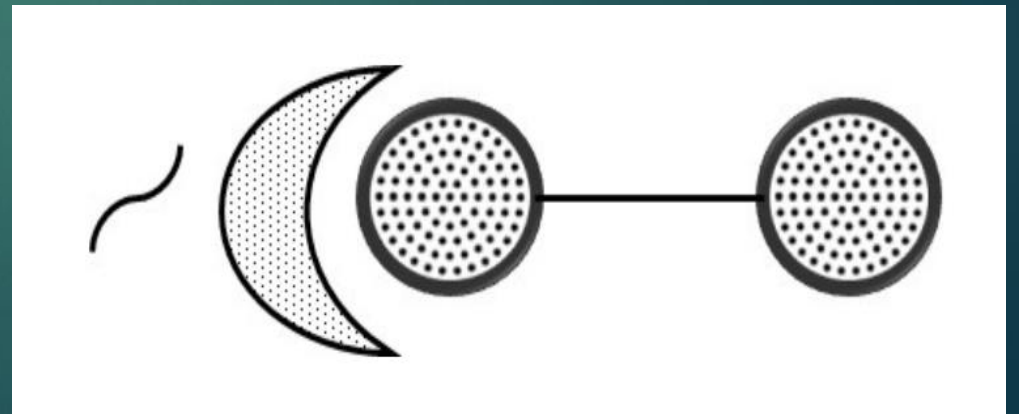
- A cluster is a set of objects such that an object in a cluster is closer (more similar) to the “center” of a cluster, than to the center of any other cluster
- The center of a cluster is often a **centroid**, the average of all the points in the cluster, or a **medoid**, the most “representative” point of a cluster



4 center-based clusters

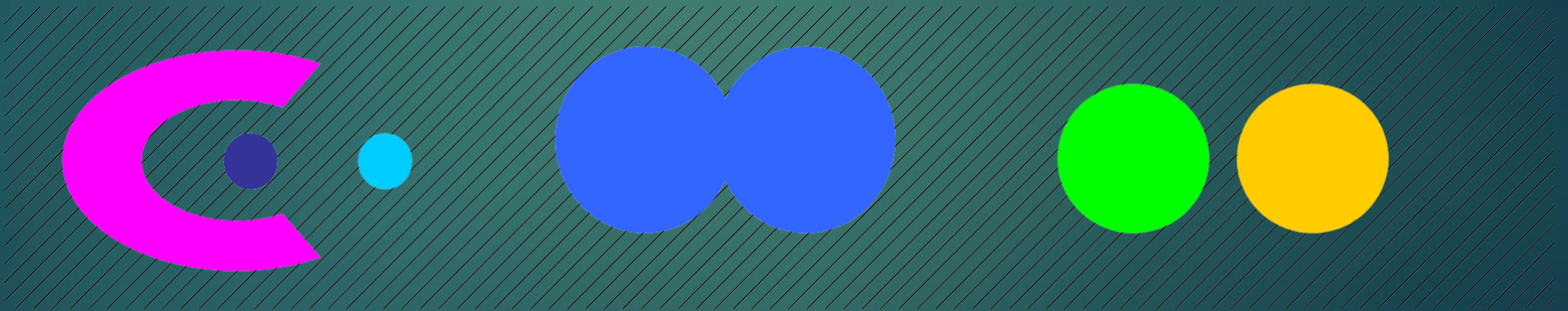
Graph-Based clusters

- If the data is represented as a graph, then a cluster can be defined as a connected component; i.e., a group of objects that are connected to one another, but that have no connection to objects outside the group
- Example: contiguity-based clusters, where two objects are connected only if they are within a specified distance of each other
- This implies that each object in a contiguity-based cluster is closer to some other object in the cluster than to any point in a different cluster



Density-Based clusters

- A cluster is a dense region of objects that is surrounded by a region of low density
- Density-based definition of a cluster is often employed when the clusters are irregular or intertwined, and when noise and outliers are present



Shared-Property (Conceptual) Clusters

- We can define a cluster as a set of objects that share some property

Types of Clusters: Conceptual Clusters

- Finds clusters that share some common property or represent a particular concept.



2 Overlapping Circles